“Applications and Techniques of MB”

DNA Extraction, Amplification & Visualizing

SGL - 7

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Molecular Biology :

Applications of molecular biology in medicine:

Most important applications are:

- Understanding the cause and mechanisms of disease
- Diagnosis, treatment and control of Disease
- Forensic medicine or medico-legal practice
- Pre-implantation genetic testing and diagnosis in embryos
- Population genetics.
- Study the AMR
- Deciphering and utilizing information written in the alphabets of life

Molecular Biology :

Extraction and purification of nucleic acids:
DNA and/or RNA Isolation

- DNA isolation is a process of purification of DNA and/or RNA from sample using a combination of physical and chemical methods.

- The first isolation of DNA was done in 1869 by Friedrich Miescher, Currently it is a routine procedure in molecular biology or forensic analysis.

- The first step for most of the molecular biology techniques procedures are extract the ***DNA*** (or for some purposes ***RNA***) from the cells and viruses and to purify it by separating it from other cellular components.

- An efficient method of nucleic acid extraction that produces ***"Purified & High-quality"*** nucleic acid product is crucial to the success of PCR, Sequencing, and many other molecular techniques.

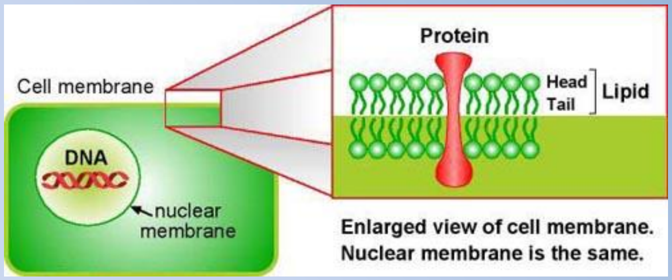
- Many methods of nucleic acid extraction have been developed, and the specific method of choice depends on the type of tissue source, as well as the intended subsequent use of the purified nucleic acid.

Molecular Biology :

Extraction and purification of nucleic acids: DNA and/or RNA Isolation

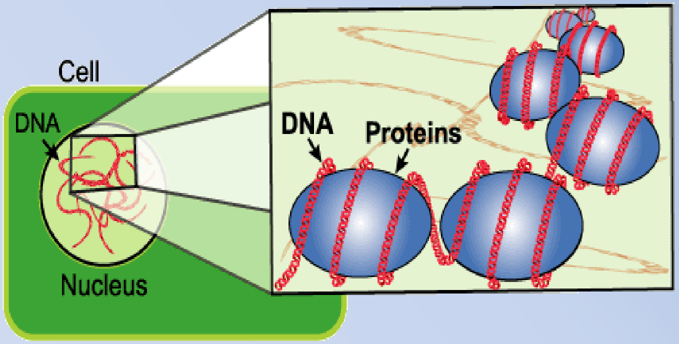
The fundamental considerations of extraction and purification of nucleic acids:

a) Effective disruption of cells or tissue ***"Cell lysis"***
b) Denaturation of nucleoprotein complexes.
c) Inactivation of endogenous ribonuclease (RNase) activity.
d) Purification of RNA & RNA away together or separately from contaminating cell components and proteins:
   - **DNA precipitation**
   - **DNA washing**
   - **DNA rehydration.**



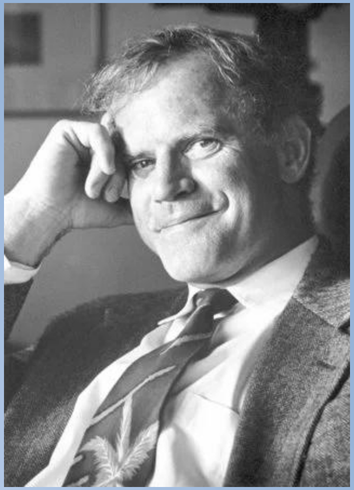
To isolate DNA, all cell and contaminant Must be Removed:

a) Cell membrane
b) Cell wall (plant, bacteria & fungi)
c) Cytoplasm and all its content
d) Nuclear membrane (animal & plant)
e) Proteins, polysaccharides & lipids



Molecular Biology :

Polymerase Chain Reaction (PCR):



- In 1993, Kary B. Mullis received the Nobel Prize in Chemistry for the invention of the polymerase chain reaction (PCR) method.

- *Thermophilus aquaticus* “Resources of ***Taq Polymerase***” is the organism that makes PCR (polymerase chain reaction) possible.
- It is an 'thermophile', capable of living in high temperatures, specifically at temperatures over 70 °C .
- It was discovered in 1969, at a time when biologists assumed that no living thing could survive at temperatures over 55 °C.

- While Thermus can 'only' withstand temperatures up to 80 °C, other organisms can live at temperatures even closer to the boiling point of water

Molecular Biology :

Polymerase Chain Reaction (PCR):

General concept of PCR:

- The polymerase chain reaction (PCR) is the one of the most powerful techniques that has been developed recently in the area of recombinant DNA research.
- PCR is a cell-free, rapid, and sensitive method for cloning DNA fragments. A standard reaction.
- Standard PCR is an in vitro procedure for amplifying defined target DNA sequences, even from very small amounts of material or material of ancient origin.
- Selective amplification requires some prior information about DNA sequences flanking the target DNA.
- Based on this information, two oligonucleotide primers of about 15–25 base pairs length are designed.
- The primers are complementary to sequences outside the 3’ ends of the target site and bind specifically to these.

Molecular Biology :

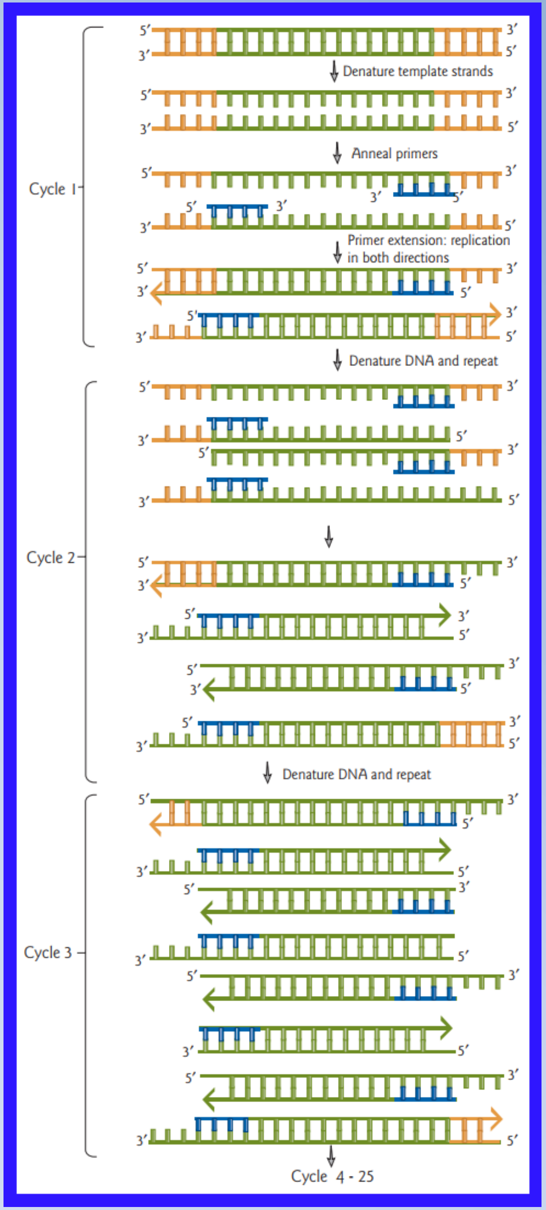
Polymerase Chain Reaction (PCR):

PCR is an *in vitro* DNA replication method. The starting material is a double stranded DNA.

The target sequence to be amplified is indicated in green.

A set of Reverse and Forward primers (blue) are added, each complementary to the end of target region.

DNA polymerase (e.g. Taq polymerase) and Deoxynucleoside triphosphates (dNTPs) are also added.

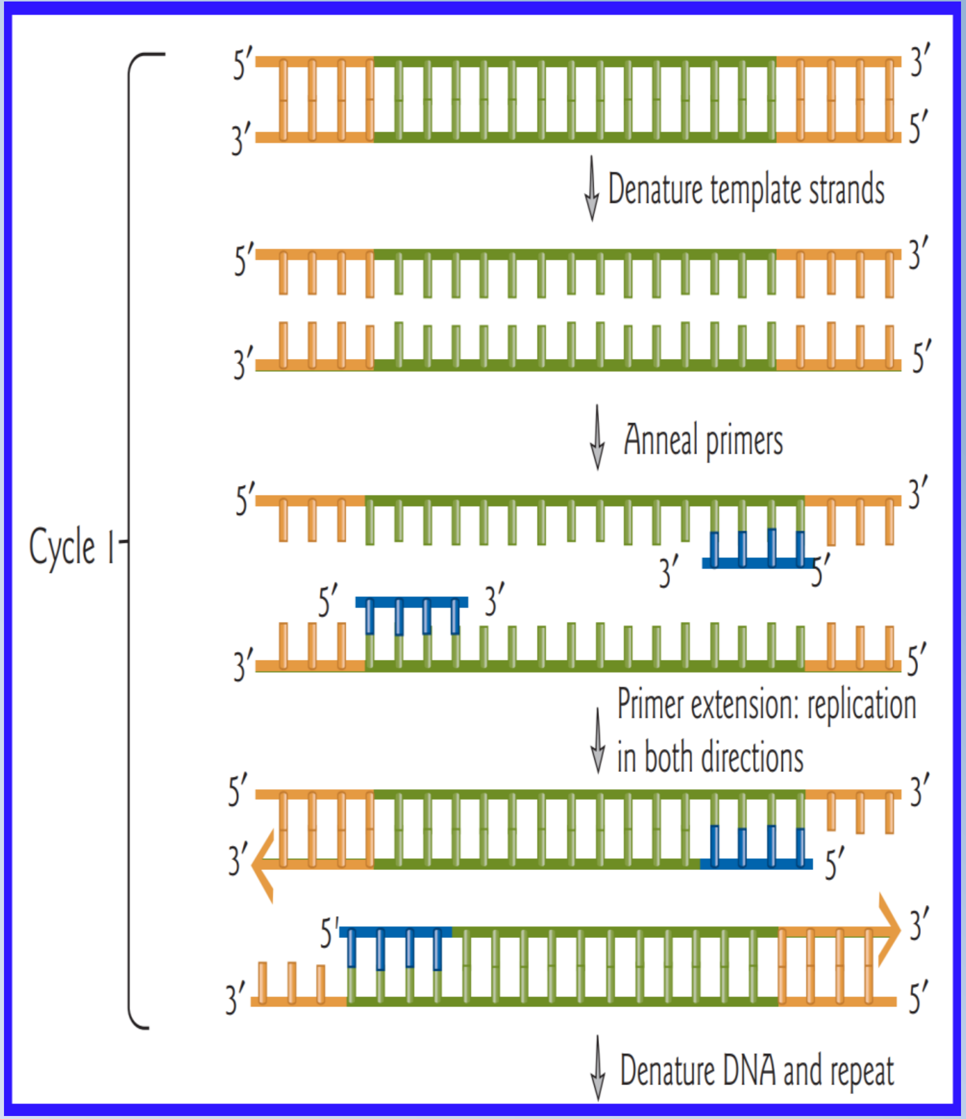


Molecular Biology :

Polymerase Chain Reaction (PCR):

Cycle 1:

1. Denaturation: Denatures the double-stranded DNA Heating to 95°C.
2. Annealing: Primers to anneal to their complementary sequences in the target DNA. Cooling to 55–65°C
3. Extension “Elongation”: extend to the end of the template strands extension is performed at 72°C

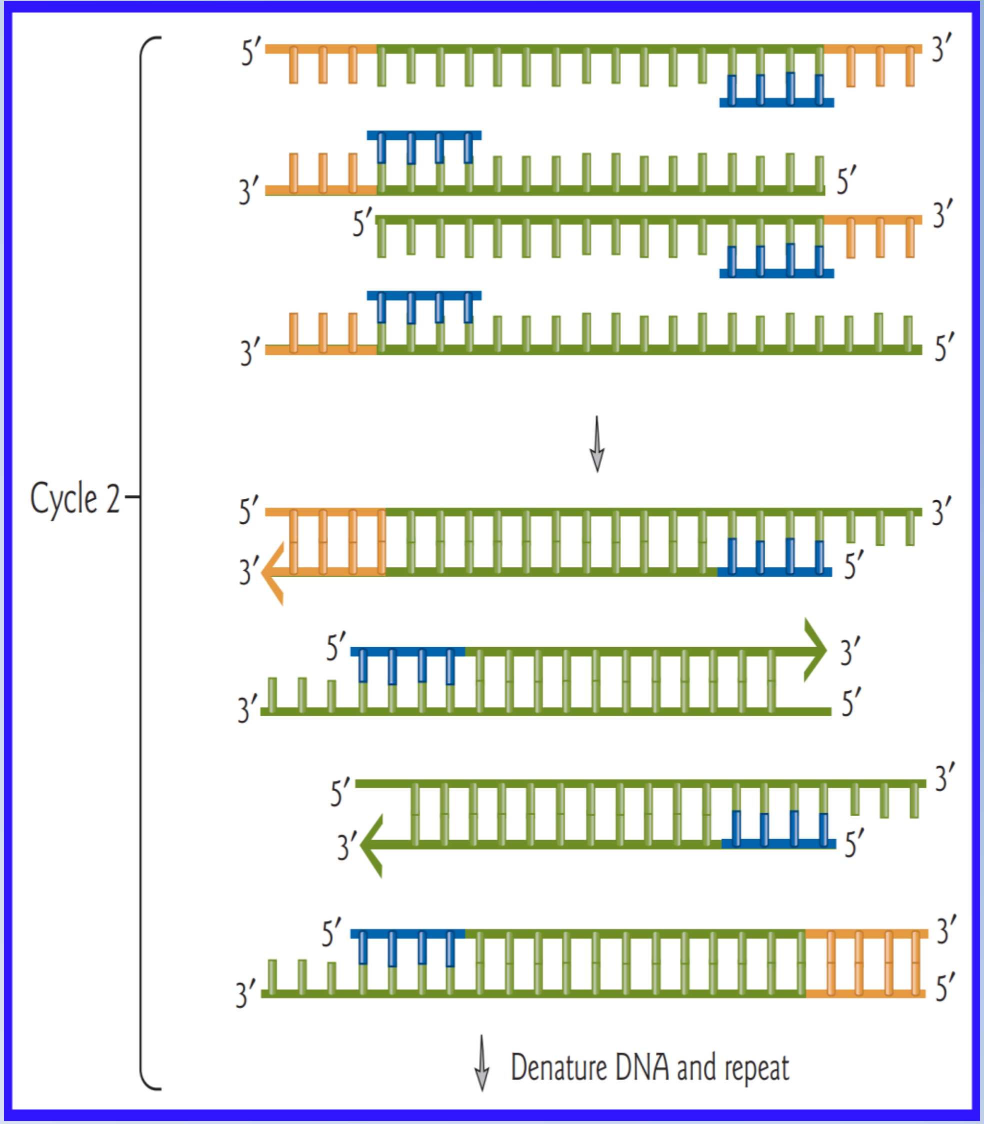


Molecular Biology :

Polymerase Chain Reaction (PCR):

Cycle 2:

1. Denaturation: Original and newly made DNA strands are denatured at 95°C.
2. Annealing: Primers to anneal to their complementary sequences in the target DNA. Cooling to 55–65°C
3. Extension “Elongation”: extend to the end of the template strands extension is performed at 72°C



Molecular Biology :

Polymerase Chain Reaction (PCR):

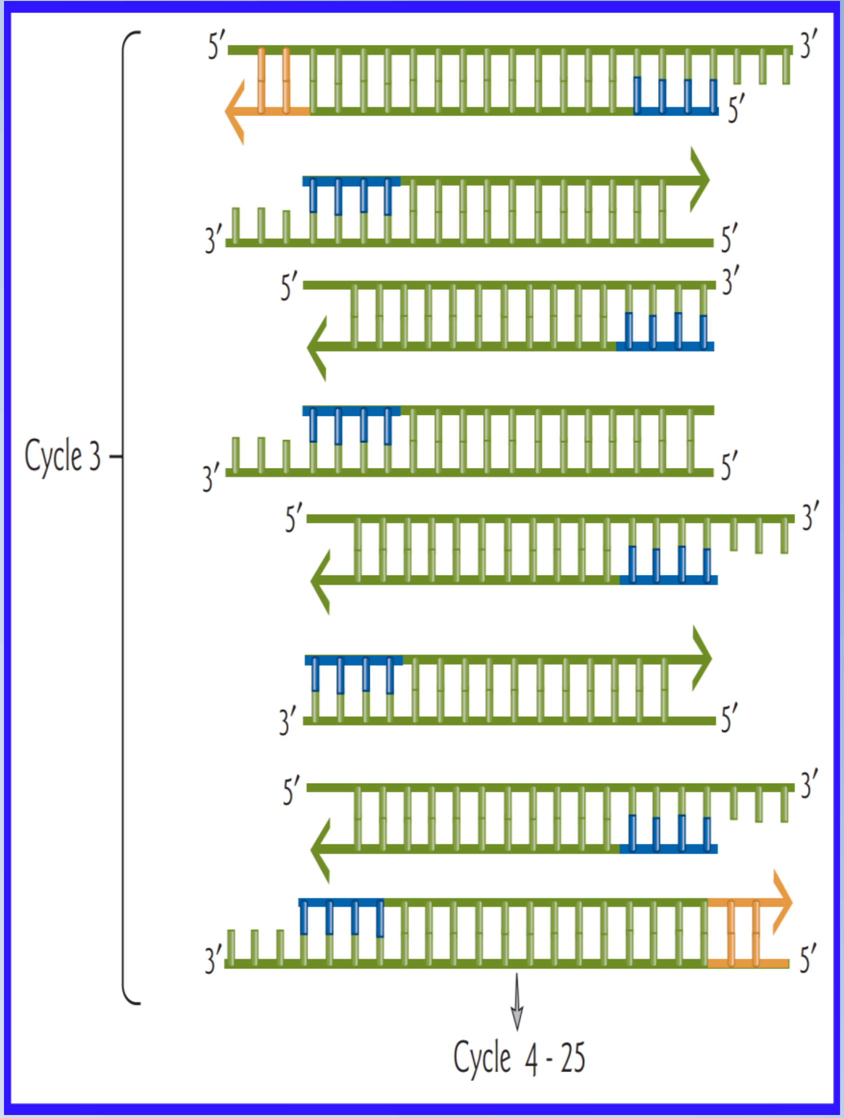
Cycle 3:

1. Denaturation: Original and newly made DNA strands are denatured at 95°C.
2. Annealing:
3. Extension:

In the third cycle, two double strand DNA molecules are generated exactly equal to the target sequence.

These two are doubled in the fourth cycle and are doubled again with each successive cycle.

These three steps are repeated from 28 to 35 times. With each cycle, more and more fragments are generated with just the region between the primers amplified.



Molecular Biology :

Polymerase Chain Reaction (PCR):

Molecular Biology :

Visualizing DNA / Nucleic Acid Probing:

- DNA molecules are small, and the information contained in their sequence is invisible.

- How does the isolate of a particular fragment of DNA can be determined, what organism it is from, what its sequence is, or what its function is?

- One method to identify the by presence of a certain DNA sequence artificially constructed pieces of DNA called probes.

- Probes can be used to identify different bacterial species in the environment and many DNA probes are now available to detect pathogens clinically.

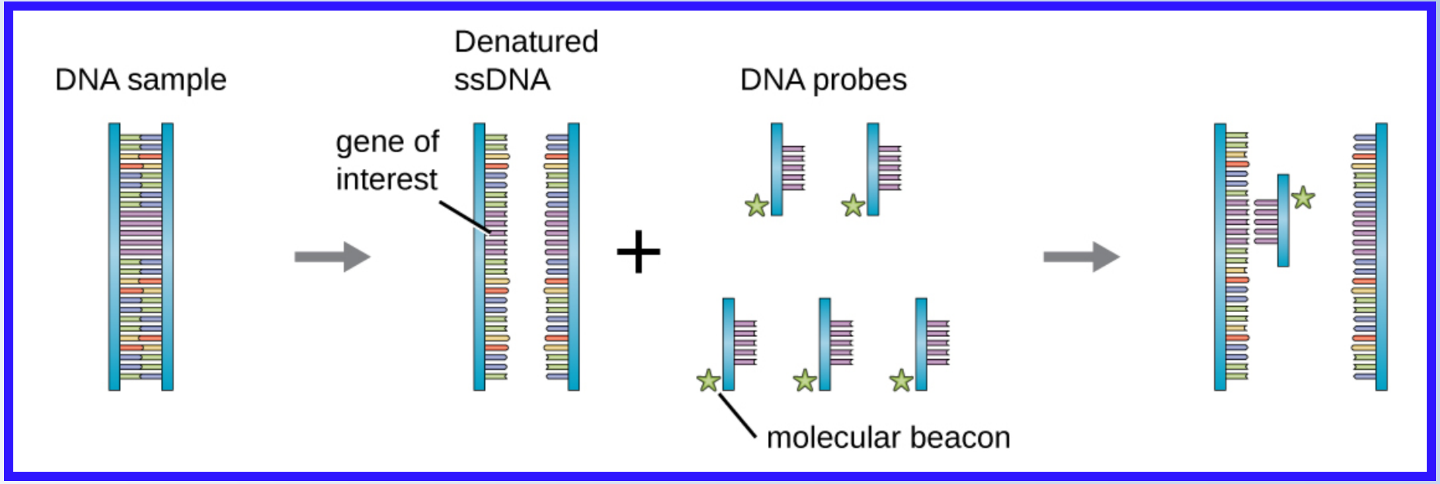
Molecular Biology :

Visualizing DNA / Nucleic Acid Probing:

- The probe is a single-stranded DNA fragment that is complementary to part of the gene of interest and different from other DNA sequences in the sample.

- The DNA probe must be labeled with a molecular tag, such as a radioactive phosphorus atom or a fluorescent dye, so that the probe and the DNA it binds to can be seen.

- The DNA sample being probed must also be denatured to make it single-stranded so that the single-stranded DNA probe can anneal to the single-stranded DNA sample at locations where their sequences are complementary.



Molecular Biology :

Visualizing DNA / Nucleic Acid Probing:

- Gel electrophoresis is a technique commonly used to separate biological molecules based on size and biochemical characteristics, such as charge and polarity.
- Agarose gel electrophoresis is widely used to separate DNA (or RNA) of varying sizes that may be generated by restriction enzyme digestion or by other means, such as the PCR.

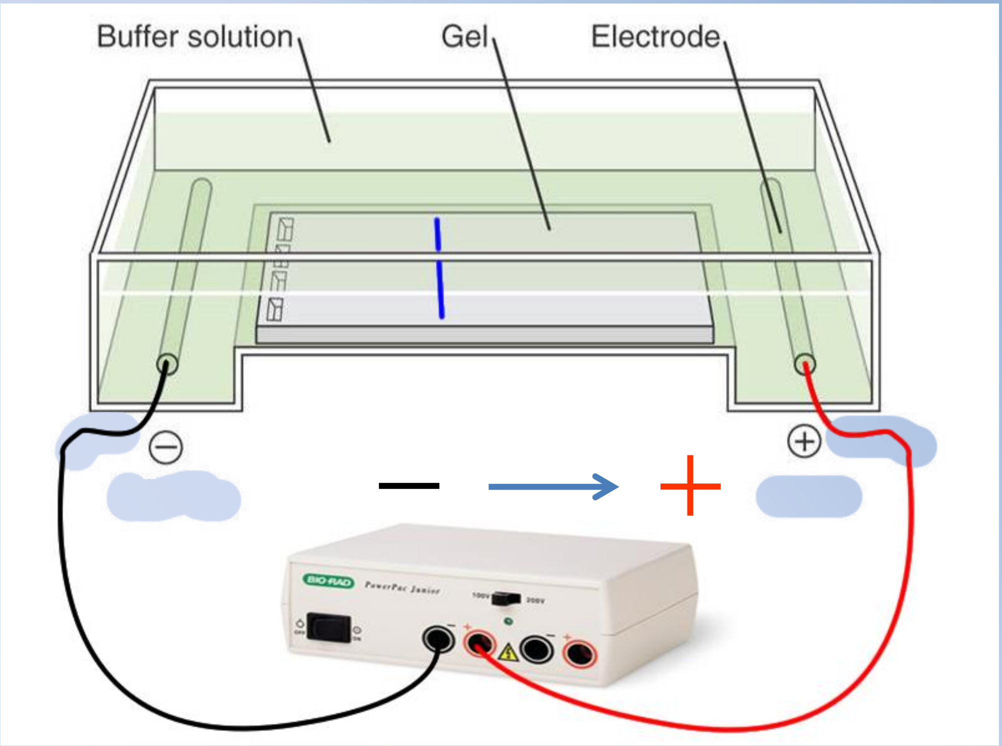
Purposes for Agarose Gel Electrophoresis:

1. Analysis of DNA size and conformation.
2. Separation and extraction of DNA fragments.
3. DNA Quantitation.
4. DNA degradation.
1. Analytical separation of PCR products.
5. Detection of Mutations or Sequence Variations.

Molecular Biology :

Visualizing DNA / Nucleic Acid Probing:

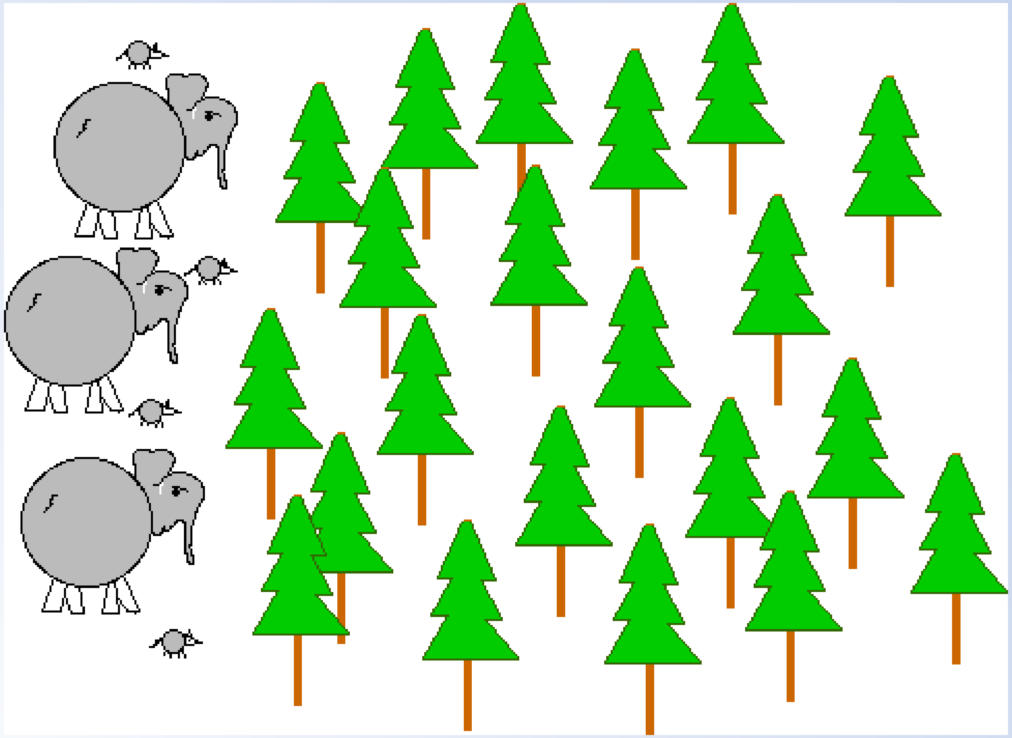
- Due to its negatively charged backbone, DNA is strongly attracted to a positive electrode.
- In agarose gel electrophoresis, the gel is oriented horizontally in a buffer solution.
- Samples are loaded into sample wells on the side of the gel closest to the negative electrode, then drawn through the agarose matrix toward the positive electrode.

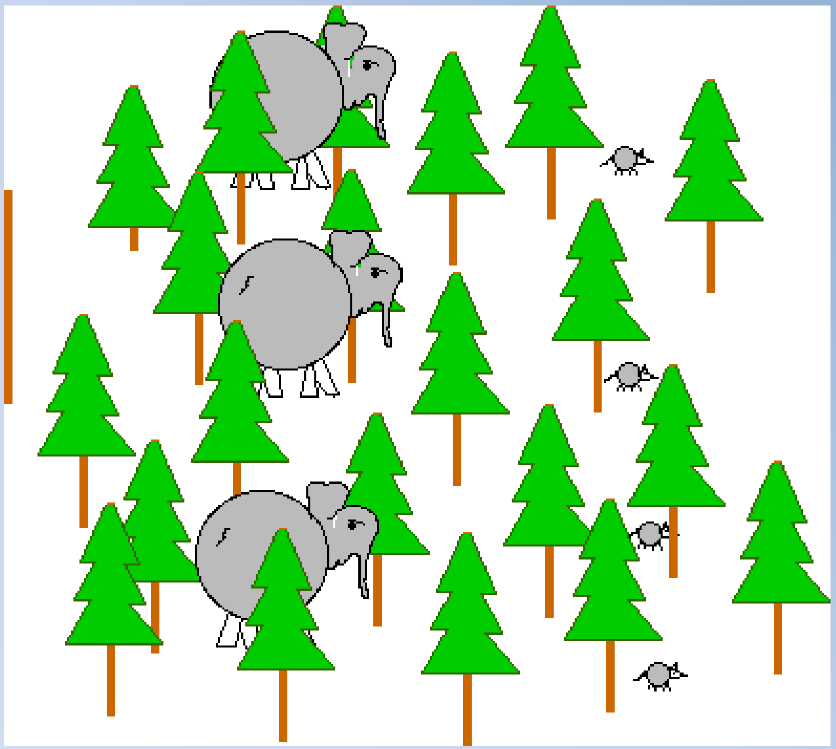


Molecular Biology :

Visualizing DNA / Nucleic Acid Probing:

- The agarose matrix impedes the movement of larger molecules through the gel, whereas smaller molecules pass through more readily.
- The distance of migration is inversely correlated to the size of the DNA fragment, with smaller fragments traveling a longer distance through the gel in same period of time.



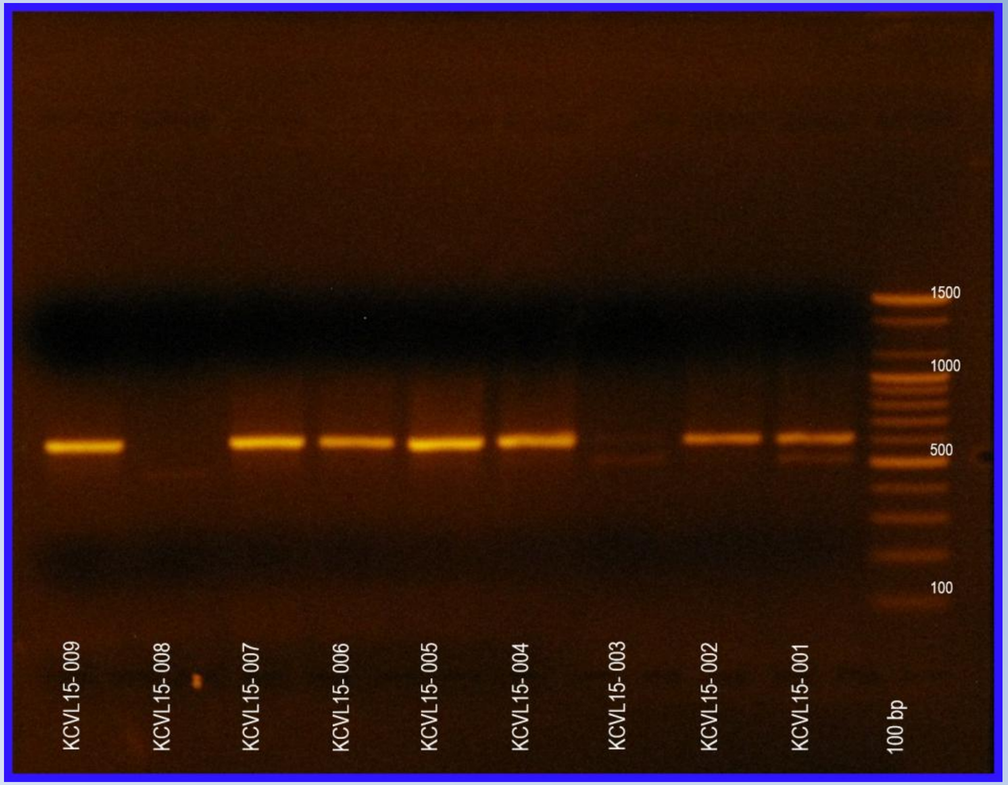


Molecular Biology :

Visualizing DNA / Nucleic Acid Probing:

- Sizes of DNA fragments within a sample can be estimated by comparison to fragments of known size in a DNA ladder also run on the same gel.

- The locations of the DNA or RNA fragments in the gel can be detected by various methods.



- One common method is adding ethidium bromide, a stain that inserts into the nucleic acids at non-specific locations and can be visualized when exposed to ultraviolet light.

- Other stains that are safer than ethidium bromide, a potential carcinogen, are now available.